Describing Motion (p251-253)

Distance and Displacement

1. What is **distance** a measure of?

2. What does **displacement** measure?

3. Distance is a **scalar quantity**. What does this mean?

4. Displacement is a **vector quantity**. What does this mean?

5. Describe the difference between distance and displacement. Draw a diagram to help explain your answer.

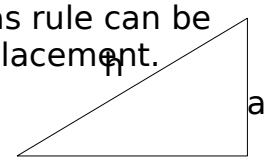
SI Units

1. Why do we use a standard set of units?

2. What are the standard units and symbols for
   i. Length
   ii. Mass
   iii. Time
   iv. Temperature

3. What is the SI unit for speed?

Hint: Pythagoras rule can be used to find displacement.
h2 = a2 + b2



Distance and Displacement Questions

Draw diagrams to work out your answers.

1. John walks north for 3m from point A to point B. He then walks 4m east to point C.
   i. What distance has he walked?
   ii. What is his displacement from point A to point C?

B C

A

2. A person walks 100m east, then 20m west. What is the
   i. Distance travelled?
   ii. Displacement?

3. A car travels 40km north, then turns around and travels another 40km south. What is
   i. The distance it has travelled?
   ii. Its displacement?

4. An ambitious hiker walks 25km west and then 35km south in a day.
   i. What is the distance the hiker walked?
   ii. Find the magnitude and direction of the hiker's displacement.

5. A car drives 110km northeast, the turns south and drives for another 45km. Calculate the displacement of the car from its starting point.

6. A squirrel leaves its nest to search for nuts, going 80m before finding some. The squirrel collects a few nuts in its mouth and heads back to its nest for 35m before it sees a cat. The squirrel drops its nuts and runs to a tree 200m in the opposite direction to the nest. When the cat has gone, the squirrel returned to the spot where it dropped the nuts.

   i. Construct a diagram of the squirrel's movements. Label each movement a, b, etc.

   ii. What was the total distance covered by the squirrel?

   iii. What is the displacement of the squirrel?

Converting Units

1. A car's speed is usually measured in km/h however the standard unit for speed is m/s. Draw a simple diagram to show how speeds can be converted between km/h and m/s.

2. Convert the following speeds into km/h. Show your working.

   i. 4 m/s

   ii. 12.5 m/s

   iii. 15 m/s

   iv. 30 m/s

3. Convert the following speeds into m/s. Show your working.

   i. 72 km/h

   ii. 36 km/h

   iii. 90 km/h

   iv. 28.8 km/h

Speed and Velocity

1. What is speed?

2. Write the word and symbol equations to calculate speed. State what each symbol means.

3. Write the rearranged formula for speed to calculate distance.

4. How is velocity different to speed?

5. A person walks 10m in 5 seconds. Calculate their speed in m/s. Show your working.

6. Barry goes for a bike ride. He travels 150km in 2 hours. What is his speed in km/h? Show your working

7. A truck travels along a road with a constant speed of 40m/s. How far has the truck travelled after 20 seconds? Show your working.